A CANONICAL MAP FROM THE DERIVED TENSOR PRODUCT OF AN INTERNAL HOM WITH ITS FIRST VARIABLE INTO THE INTERNAL HOM OF THE INTERNAL HOMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the choice of K-injective representatives for M and L and of a K-flat representative for K, and the definition of the morphism through the totalized evaluation map together with the identification of the two sides.

Editorial note. The label lemma-internal-hom-evaluate names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-internal-hom-evaluate –start 8964 –end 8995 –local-id 3571 returned RESULT=OK (L3571 free, occupancy FREE, public interval clear). Candidate taken from operations/pool-cards/stacks-sites-cohomology-halfB.json.

Editorial note. The proof ends with the sentence 'We omit the proof that this is functorial in all three objects of D(O).' — the omitted verification is the functoriality of a map that is constructed explicitly, not a mathematical step of the argument, so the statement is counted and the sentence is recorded here verbatim.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The evaluation map for complexes of sheaves of modules. Same chapter, the lemma constructing the map from the total tensor product of the sheaf hom complex with a complex to the sheaf hom complex of the sheaf homs. Supplies the underlying map of complexes whose totalization gives the canonical morphism of the statement.
- Existence of K-injective and K-flat representatives. Same chapter, the lemmas producing K-injective representatives of bounded below complexes and K-flat representatives of arbitrary complexes. Supply the choice of complexes I, J and K so that the two sides of the displayed map compute the two derived functors.

Source reference. Original reference 0A60: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8700–8700. Prerequisite pointer invoked at proof

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line(s) 8989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let K, L, M be objects of $D(\mathcal{O})$. There is a canonical morphism*

$$R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}} K \longrightarrow R\mathcal{H}om(R\mathcal{H}om(K, L), M)$$

in $D(\mathcal{O})$ functorial in K, L, M.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing M, a K-injective complex $\mathcal{J}^\bullet$ representing L, and a K-flat complex $\mathcal{K}^\bullet$ representing K. The map is defined using the map

$$\mathrm{Tot}(\mathcal{H}om^\bullet(\mathcal{J}^\bullet, \mathcal{I}^\bullet) \otimes_{\mathcal{O}} \mathcal{K}^\bullet) \longrightarrow \mathcal{H}om^\bullet(\mathcal{H}om^\bullet(\mathcal{K}^\bullet, \mathcal{J}^\bullet), \mathcal{I}^\bullet)$$

of Lemma 0A60. By our particular choice of complexes the left hand side represents $R\mathcal{H}om(L, M) \otimes^{\mathbf{L}}_{\mathcal{O}} K$ and the right hand side represents $R\mathcal{H}om(R\mathcal{H}om(K, L), M)$. We omit the proof that this is functorial in all three objects of $D(\mathcal{O})$. □